

QA AUTOMATION CAPSTONE: PROJECT DESCRIPTION

PROJECT NAME

Enterprise E-Commerce UI Automation Framework using Playwright and Java Script.

PROJECT OVERVIEW

The Enterprise E-Commerce UI Automation Framework using Playwright and JavaScript is designed to validate the most critical workflows of the NIKKA website. It covers authentication, product search, cart, address and shipping, payment and checkout, order history, API services, user profile, and customer support. The framework emphasizes scalability, resiliency, and maintainability, ensuring seamless execution across Chromium, Firefox, and WebKit. By integrating UI automation, API validation, visual regression, and AI-assisted test generation, it delivers enterprise-grade reliability, performance, and reporting suitable for production environments.

WEBSITE NAME

NIKKA

WEBSITE URL

<https://www.nykaa.com/>

PROJECT OBJECTIVES

- Prepare a complete test plan for the selected website.
- Create a minimum of 120 test cases across 8 services.
- Cover positive, negative, validation, navigation, and regression scenarios.
- Automate the planned tests using Playwright with JavaScript.
- Generate execution evidence through test reports, screenshots, and traces wherever required.
- Present the project as a QA/testing capstone, not as a development project.

TEST CASE DISTRIBUTION

- **Services:** 8
- **Test cases per service:** ~15
- **Total per browser:** 120
- **Browsers:** Chromium, Firefox, WebKit
- **Grand Total Executions:** 360

PLANNED SERVICES

S.NO	Service / Module	Example Testing Coverage	Planned Test Cases
01.	Authentication & User Accounts	Valid/invalid login, logout, password reset, account creation, duplicate email, session timeout, lockout, remember-me, protected page access	15
02.	Product Search & Discovery	Exact/partial search, no results, filters (price, category, rating), sorting (price, relevance, newest), pagination, empty input, performance validation	15
03.	Cart Management	Add/remove/update items, multiple products, invalid quantity, max quantity, subtotal/tax/total calculation, coupon handling, persistence, cart icon count, cart clears after checkout	15
04.	Address & Shipping	Add/edit/delete address, mandatory fields validation, invalid pin code, multiple addresses, default selection, shipping method (standard/express), charges calculation, checkout summary	15
05.	Payment & Checkout	Guest/registered checkout, valid/invalid card, expired card, CVV validation, insufficient balance (mock), multiple payment methods, coupon code, payment retry, timeout simulation, confirmation page	15
06.	Order History & Tracking	View past orders, validate details, track status, cancel order, refund request, re-order, empty history, filtering (date/status), sorting (newest/oldest), cancellation reflected in history	15
07.	API Service (Mocking & Backend Validation)	Mock success/failure, delayed response, empty/partial/duplicate data, unauthorized access, schema validation, error messages, retry mechanism, caching, large payload handling	15
08.	User Profile Management	View/edit profile, mandatory fields validation, invalid email/phone, change password (valid/invalid), upload/change profile picture, invalid format/size, notification preferences, delete account, persistence	15

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CONCLUSION

This capstone project establishes a future-ready automation solution that balances technical innovation with business impact. By validating end-to-end customer journeys and integrating AI, API resilience, and enterprise reporting, it enhances trust, usability, and operational stability. The framework positions NIKKA as a benchmark in modern quality engineering, showcasing best practices in scalable automation and intelligent test design.